



ML ENDGAME



When Thanos and Darkseid, two of the universe's deadliest villains, join forces to conquer Earth, they bring with them the full power of the Infinity Gauntlet and the Anti-Life Equation. With no single hero or team able to take on this apocalyptic threat, Earth's greatest defenders must unite. The Avengers [Marvel] and Justice League [DC] must put aside their differences to form a powerful alliance. But there's a catch—this is not about intuition or mere strategy. As a machine learning student, you've been tasked with the challenge of forming the perfect teams using data-driven insights, with the only tool at your disposal: the K-Nearest Neighbors (KNN) algorithm.

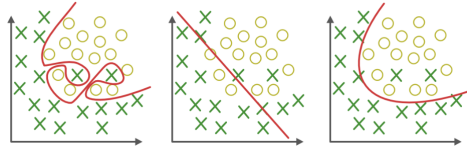
<i>Avengers</i>			<i>Justice League</i>		
Character	Embedding	Popularity	Character	Embedding	Popularity
Iron Man	[3.8, 3.7, 4.5, 3.6]	95	Superman	[4.6, 4.9, 4.2, 4.8]	98
Captain America	[4.5, 3.8, 4.0, 4.3]	90	Batman	[3.9, 3.8, 4.3, 4.2]	97
Thor	[4.5, 4.9, 4.3]	92	Wonder Woman	[4.8, 4.5]	93
Hulk	[4.9, 4.9, 4.2]	88	Flash	[4.9, 4.4]	89
Black Panther	[4.4, 4.8, 3.9, 2.7, 4.5]	87	Aquaman	[2.6, 4.4, 4.1, 4.5]	86
Doctor Strange	[4.9, 4.6]	89	Green Lantern	[4.1, 4.8, 4.5]	85
Spider-Man	[4.6, 4.9]	96	Cyborg	[4.6, 3.9, 3.8]	84

Answer following questions:

• Beginner

(1 Marks each)

- Cyborg can only sense energy/information from d -dim, hence his fighting capacity is equal to VC-dim. Which of the following statements is true?
(a) $d_{VC} = d + 1$ (b) $d_{VC} \geq d + 1$ (c) $d_{VC} \leq d + 1$ (d) All of the above
- In order to win, your KNN model needs to be consistent, and this happens if it produces a hypothesis that perfectly fits the...
(a) cross-validation data (b) overall dataset (c) test data (d) training data
- We know you have practiced LR more instead of KNN, where the hypothesis is given by $h(x) = b_0 + b_1x$. What will be the value of $h(x)$ when x is 0? Explain.
- In a K-Nearest Neighbors (KNN) model applied to superhero collaboration prediction, the value of **K** is increased. What effect is most likely to occur for model?
(a) Sensitive to noise in the training data increases. (b) Less sensitive to noise but risks underfitting.
(c) Model computational complexity decreases. (d) Distance metric becomes irrelevant.
- In this team-up you can either have them all wear the same sized armor (Min-Max Normalization) or have each hero's armor tailored to fit their unique abilities (Standardization). Explain which of these two mentioned methods you will use for data pre-processing of KNN model and why?
- Assume Thanos army has Positive (Powerful than Avengers) and Negative warriors with a sensitivity of being powerful 0.90 and specificity of 0.80. A single test is applied to each army member in which the powerful population was 30%. What is the probability that a member negative for this test has no power? Explain your reasoning.
(a) Less than 50% (b) 70% (c) 95% (d) 65%
- So, as usual, you decided to ask ChatGPT, which proposes three classifiers, as shown below. Which of the following is/are TRUE

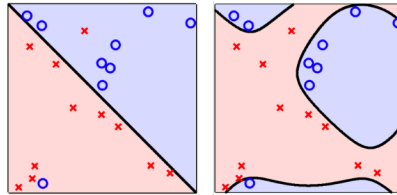


- (a) Left high robustness, poor fit.
 (b) Left has poor robustness, high fit.
 (c) Middle has poor robustness, high fit.
 (d) Right has high robustness, poor fit.

• Intermediate

(2 Marks each)

- You created combinations and manually labeled data as win/lose. Then you decided to use non-linear transformation of data.
 - Why does non-linear transformation of data sometimes improve the performance of Linear models?
 - Below, you are given decision boundaries of logistic regression (Left) with original data and KNN (Right) with non-linear transformation of data. Which one would you prefer and why?



- In terms of bias and/or variance of your model: Which aspect of your model will be fixed if the following is done
 - Get more training data
 - Try a smaller set of features
 - Try adding more features
 - Try decreasing regularization constant λ
- Your KNN failed, and Thanos wiped the half of humanity! Meanwhile, in the future, to fix this, Hulk and Ironman are trying to stabilise the Infinity Gauntlet using GLM trained on the distribution of radiation signatures of each stone. Help them by matching the link function to the corresponding correct distribution

Distribution	Link Function $\mathbf{Xb} = g(\mu)$
Normal	
Exponential	
Gamma	$g(\mu) \in [\ln(\frac{\mu}{1-\mu}), \mu, \mu^{-1}, \mu^{-2}, \ln \mu]$
Bernoulli	
Categorical	

• Advanced

- You used your good friend ChatGPT for making clusters of similar superheroes; for $K = 3$, it shows Ironman, Batman and Spiderman as similar. Verify this claim using the following steps: **(3 Marks)**
 - Find the pairwise Euclidean distance d between all three superheroes.
 - Make an adjacency matrix of Gaussian similarity ($\exp(-\frac{d^2}{2})$).
 - Calculate the error for the ChatGPT hypothesis based on the computed similarities from above.
- (Climax!) So don't neglect the input from the dude that can see the future [Doctor Strange]. Either for a smaller or higher value of K , you are bound to lose the battle. Explain it in terms of bias-variance tradeoff for KNN algorithm? **(2 Marks)**
- The Man, The Myth, The Legend. OG Strategist [Batman] strategies and says the longer you plan, the harder it is to win the battle as Darkseid and Thanos are summoning their armies. Find the time-complexity of the KNN Algorithm and K-fold KNN Algorithm so the team can win faster by choosing the best implementation. **(2 Marks)**



Machine Learning (CSE543/ECE543)

Duration 120 min



ML ENDGAME

When Thanos and Darkseid, two of the universe's deadliest villains, join forces to conquer Earth, they bring with them the full power of the Infinity Gauntlet and the Anti-Life Equation. With no single hero or team able to take on this apocalyptic threat, Earth's greatest defenders must unite. The Avengers [Marvel] and Justice League [DC] must put aside their differences to form a powerful alliance. But there's a catch—this is not about intuition or mere strategy. As a machine learning student, you've been tasked with the challenge of forming the perfect teams using data-driven insights, with the only tool at your disposal: the Support Vector Machines (SVM).



<i>Avengers</i>			<i>Justice League</i>		
Character	Embedding	Popularity	Character	Embedding	Popularity
Iron Man	[3.8, 3.7, 4.5, 3.6]	95	Superman	[4.6, 4.9, 4.2, 4.8]	98
Captain America	[4.5, 3.8, 4.0, 4.3]	90	Batman	[3.9, 3.8, 4.3, 4.2]	97
Thor	[4.5, 4.9, 4.3]	92	Wonder Woman	[4.8, 4.5]	93
Hulk	[4.9, 4.9, 4.2]	88	Flash	[4.9, 4.4]	89
Black Panther	[4.4, 4.8, 3.9, 2.7, 4.5]	87	Aquaman	[2.6, 4.4, 4.1, 4.5]	86
Doctor Strange	[4.9, 4.6]	89	Green Lantern	[4.1, 4.8, 4.5]	85
Spider-Man	[4.6, 4.9]	96	Cyborg	[4.6, 3.9, 3.8]	84

Answer following questions:

• Beginner

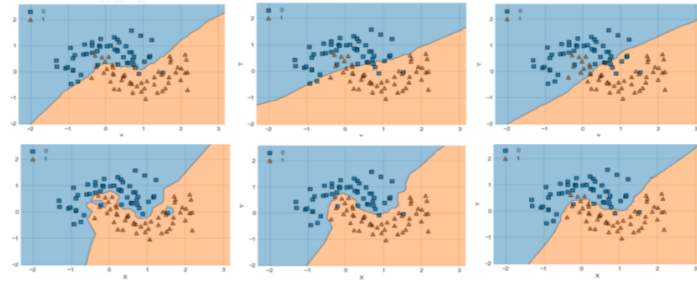
1 Mark each

- Cyborg can only sense energy/information from d -dim, hence his fighting capacity is equal to VC-dim. In order to prove that $d_{VC} \leq d + 1$, we have to show
 - There are $d + 1$ points we cannot shatter
 - There are $d + 2$ points we cannot shatter
 - We cannot shatter any set of $d + 1$ points
 - We cannot shatter any set of $d + 2$ points
- Apart from win/lose there can be other scenarios. Help Superman by stating two scalable strategies to extend a binary classifier to multi-class classification. Note you can't change either the model type or the loss function.
- In order to win, your KNN model needs to be consistent, and this happens if it produces a hypothesis that perfectly fits the...
 - cross-validation data
 - overall dataset
 - test data
 - training data
- We know you have practiced LR more instead of KNN, where the hypothesis is given by $h(x) = b_0 + b_1x$. What will be the value of $h(x)$ when x is 0? Explain.
- Explain the difference between stratified and random K-fold cross-validation strategy.
- In this team-up you can either have them all wear the same sized armor (Min-Max Normalization) or have each hero's armor tailored to fit their unique abilities (Standardization). Explain which of these two mentioned methods you will use for data preprocessing of RBF Kernelized SVMs and why?
- Assume Thanos army has Positive (Powerful than Avengers) and Negative warriors with a sensitivity of being powerful 0.90 and specificity of 0.80. A single test is applied to each army member in which the powerful population was 30%. What is the probability that a member negative for this test has no power? Explain your reasoning.
 - Less than 50%
 - 70%
 - 95%
 - 65%

• Intermediate

(2 Marks each)

- You created combinations and manually labeled data as win/lose. Then, you decided to use non-linear transformation of data.
 - Why does non-linear transformation of data sometimes improve the performance of Linear models?
 - Below, you are given decision boundaries of various classifiers. Which one corresponds closely to KNN with the largest value of K ?



- In terms of bias and/or variance of your model: Which aspect of your model will be fixed if the following is done
 - Get more training data
 - Try a smaller set of features
 - Try adding more features
 - Try decreasing regularization constant λ
- Your KNN failed, and Thanos wiped the half of humanity! Meanwhile, in the future, to fix this, Hulk and Ironman are trying to stabilise the Infinity Gauntlet using GLM trained on the distribution of radiation signatures of each stone. Help them by matching the link function to the corresponding correct distribution

Distribution	Link Function $\mathbf{Xb} = g(\mu)$
Normal	
Exponential	
Gamma	$g(\mu) \in [\ln(\frac{\mu}{1-\mu}), \mu, \mu^{-1}, \mu^{-2}, \ln \mu]$
Bernoulli	
Categorical	

• Advanced

- Predicting collaborations between superheroes from the Avengers and the Justice League, which situation best indicates a need for the RBF kernel instead of the linear kernel? (1 Mark)
 - Superheroes with high cosine similarity in their embeddings have conflicting collaboration labels.
 - Superheroes belong to different universes (Avengers and Justice League), requiring additional features.
 - The training data is linearly separable.
 - Increasing the regularization parameter CC improves collaboration prediction performance.
- The team of Black Panther, Aquaman, and Captain America was predicted to defeat Steppenwolf and the Parademons using an SVM with an RBF Kernel. However, they started losing due to an error in the decision function. Cyborg can help if you derive the correct function. Derive $f(\mathbf{x})$ to assist Cyborg in correcting the strategy. (2 Mark)
 - Embeddings:** Black Panther, Aquaman, and Captain America, respectively.
 - Labels:** $y = [1, 1, -1]$, **Lagrange Multipliers:** $\alpha = [0.8, 1.2, 0.5]$.
 - Decision Function:** $f(\mathbf{x}) = \sum_{i=1}^n \alpha_i y_i K(\mathbf{x}_i, \mathbf{x}) + b$.
- Assuming RBF kernel, in this team-up you can either have them all wear the same sized armor (Min-Max Normalization) or have each hero's armor tailored to fit their unique abilities (Standardization). Explain which of these two mentioned methods you will use for data preprocessing of SVC model and why? (2 Mark)
- Hulk has suggested you derive the optimization problem for soft-margin SVM, including the objective function with the penalty term C and the slack variables $\mathcal{E}$. This can make the difference in war (believe it, this giant has 7 PhDs); while he is out battling armies, you battle with the derivation. (2 Mark)
- After decimating waves of armies, Dr. Banner (Hulk), while having tacos, asked you: How does the regularization parameter C affect the decision boundary and classification of potential collaborations? He may offer you one taco if you give the correct answer. (2 Mark)